

POISSON DISTRIBUTION

3.7 POISSON DISTRIBUTION

Poisson Distribution due to French mathematician Simeon Denis Poisson (1837) is a discrete probability distribution.

S. D. Poisson introduced Poisson distribution as a rare distribution of rare events i.e., the events whose probability of occurrence is very small but the number of trials which could lead to the occurrence of the event, are very large.

Suppose that, in a large number of parts manufactured by a machine, samples of 100 parts are taken from a product which is 2% defective. Then the probabilities of 0, 1, 2, ..., 99, 100 defectives in a single sample are given by the successive terms of the Binomial distribution $(0.98 + 0.02)^{100}$. Expanding the binomial in full will be a tedious process. This difficulty can be avoided by the Poisson distribution.

DERIVATION OF THE POISSON DISTRIBUTION

[JNTU 2004, 2006S, (A) Dec. 2009, (K) 2010S, (H) May 2011, (K) Nov. 2011 (Set No. 3)]

The Poisson distribution can be derived as a limiting case of the Binomial Distribution under the following conditions:

- (i) p , the probability of the occurrence of the event is very small.
- (ii) n is very very large, where n is number of trials i.e., $n \rightarrow \infty$.
- (iii) np is a finite quantity, say $np = \lambda$, then λ is called the parameter of the Poisson distribution.

In the Binomial distribution, the probability $P(r)$ or $P(X = r)$ of r successes in a series of independent trials is given by

$$\begin{aligned} P(r) &= {}^nC_r p^r q^{n-r} = {}^nC_r p^r (1-p)^{n-r} \\ &= \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \quad \dots (1) \end{aligned}$$

Put $np = \lambda$. Then $n = \frac{\lambda}{p}$

Hence (1) becomes,

$$\begin{aligned} P(r) &= \frac{\frac{\lambda}{p} \left(\frac{\lambda}{p} - 1 \right) \left(\frac{\lambda}{p} - 2 \right) \dots \left(\frac{\lambda}{p} - r + 1 \right)}{r!} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \\ &= \frac{\lambda(\lambda-p)(\lambda-2p)\dots[\lambda-(r-1)p]}{r! p^r} \cdot p^r \cdot \frac{(1-p)^n}{(1-p)^r} \\ &= \frac{\lambda(\lambda-p)(\lambda-2p)\dots[\lambda-(r-1)p]}{r!} \cdot \frac{\left(1 - \frac{\lambda}{n} \right)^n}{(1-p)^r} \quad \dots (2) \end{aligned}$$

As $n \rightarrow \infty$, $p \rightarrow 0$, so that $np = \lambda$, we have

$$\begin{aligned}
 P(r) &= \frac{\lambda \cdot \lambda \dots r \text{ factors}}{r!} \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \cdot \lim_{p \rightarrow 0} \frac{1}{(1-p)^r} \\
 &= \frac{\lambda^r}{r!} \cdot \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \left[\because \lim_{p \rightarrow 0} (1-p)^r = 1 \text{ for a given } r \right] \\
 &= \frac{\lambda^r}{r!} e^{-\lambda} \left[\because \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{\lambda}{n}\right)^{-n/\lambda} \right]^{-\lambda} = e^{-\lambda} \right]
 \end{aligned}$$

$$\therefore P(r) = \text{Probability of } r \text{ successes} = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

This is known as Poisson distribution.

Putting $r = 0, 1, 2, \dots$ etc., the probabilities of $0, 1, 2, \dots, r, \dots$ successes are given by $e^{-\lambda}, \lambda e^{-\lambda}, \frac{\lambda^2}{2!} e^{-\lambda}, \dots, \frac{\lambda^r}{r!} e^{-\lambda}, \dots$ respectively, where $\lambda (> 0)$ is a parameter.

The sum of these probabilities is unity as it should be.

3.8 DEFINITION. POISSON DISTRIBUTION

[JNTU 2006S, 2008S, (A), (H) Dec. 2009 (Set No. 4)]

A random variable X is said to follow a Poisson distribution if it assumes only non-negative values and its probability density function is given by

$$p(x, \lambda) = P(X = x) = \begin{cases} \frac{e^{-\lambda} \cdot \lambda^x}{x!}; & x = 0, 1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases} \quad \dots (1)$$

Here $\lambda > 0$ is called the parameter of the distribution.

It is a distribution suitable for rare events for which the probability of occurrence p is very small and the number of trials n is very large where np is finite. For example, consider the number of persons born blind per year in a large city. Individually, being born blind is a rare event and its probability is very small. But in a large city the total number of births every year is high. As a result, we may get some blind births.

Note : 1. It should be noted that

$$\sum_{x=0}^{\infty} P(X = x) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} \cdot e^{\lambda} = 1$$

Hence equation (1) is a probability function.

2. The Poisson Distribution function is

$$F_X(x) = P(X \leq x) = \sum_{r=0}^x p(r) = e^{-\lambda} \sum_{r=0}^x \frac{\lambda^r}{r!}, \quad x = 0, 1, 2, \dots$$

Examples of Poisson Distribution :

- (i) The number of defective electric bulbs manufactured by a reputed company.
- (ii) The number of telephone calls per minute at a switch board.
- (iii) The number of cars passing a certain point in one minute.
- (iv) The number of printing mistakes per page in a large text.
- (v) The number of particles emitted by a radio-active substance.
- (vi) The number of persons born blind per year in a large city.

Conditions of Poisson Distribution :

The Poisson Distribution is used under the following conditions :

1. The variable (number of occurrences) is a discrete variable.
2. The occurrences are rare.
3. The number of trials (n) is large.
4. The probability of success (p) is very small (very close to zero)
5. $np = \lambda$ is finite.

3.9 CONSTANTS OF THE POISSON DISTRIBUTION**1. The Mean of the Poisson Distribution**

[JNTU (H) Nov. 2009 (Set No. 4), (K) May 2012, Dec. 2013, Mar. 2014, (A) Dec. 2014]

$$\begin{aligned}
 \text{Mean} = E(X) &= \sum_{x=0}^{\infty} x \cdot p(x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \quad [\because x! = x(x-1)!] \\
 &= e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^{y+1}}{y!} \quad (\text{Putting } y = x - 1) \\
 &= \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{-\lambda} \cdot \lambda \cdot e^{\lambda} \left[\because \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{\lambda} \right] \\
 &= \lambda (= np)
 \end{aligned}$$

Thus the parameter λ is the Arithmetic Mean of the Poisson distribution.

2. Variance of Poisson Distribution

[JNTU (H) Nov. 2009 (Set No. 4), (K) May 2012, Dec. 2013, Mar. 2014 (Set No. 1), (A) Dec. 2014]

$$V(X) = E(X^2) - [E(X)]^2$$

$$= \sum_{x=0}^{\infty} x^2 p(x) - \lambda^2 \quad [\because \lambda = \text{mean of P.D.}]$$

$$= \sum_{x=0}^{\infty} x^2 \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} - \lambda^2 = e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \cdot \lambda^x}{(x-1)!} - \lambda^2$$

$$= e^{-\lambda} \sum_{x=1}^{\infty} [(x-1) + 1] \cdot \frac{\lambda^x}{(x-1)!} - \lambda^2$$

$$\begin{aligned}
&= e^{-\lambda} \left[\sum_{x=1}^{\infty} (x-1) \cdot \frac{\lambda^x}{(x-1)(x-2)!} + \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \right] - \lambda^2 \\
&= e^{-\lambda} \left[\sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} + \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \right] - \lambda^2 \\
&= e^{-\lambda} \left[\sum_{y=0}^{\infty} \frac{\lambda^{y+2}}{y!} + \sum_{z=0}^{\infty} \frac{\lambda^{z+1}}{z!} \right] - \lambda^2 \quad (\text{Putting } y = x-2, z = x-1) \\
&= e^{-\lambda} \left[\lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} + \lambda \sum_{z=0}^{\infty} \frac{\lambda^z}{z!} \right] - \lambda^2 = e^{-\lambda} [\lambda^2 \cdot e^{\lambda} + \lambda e^{\lambda}] - \lambda^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda
\end{aligned}$$

Thus variance = λ

Hence the variance of the distribution = mean of the distribution = λ

Further, Standard deviation of the Poisson distribution, $\sigma = \sqrt{\lambda}$

3. Mode of the Poisson Distribution

Mode is the value of x for which the probability $p(x)$ is maximum.

$$\therefore p(x) \geq p(x+1) \quad \text{and} \quad p(x) \geq p(x-1)$$

$$\text{Now } p(x) \geq p(x+1) \Rightarrow \frac{e^{-\lambda} \cdot \lambda^x}{x!} \geq \frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!}$$

$$\begin{aligned}
&\Rightarrow 1 \geq \frac{\lambda}{x+1} \quad \text{or} \quad \frac{\lambda}{x+1} \leq 1 \Rightarrow \lambda \leq x+1 \quad \text{or} \quad x+1 \geq \lambda \\
&\Rightarrow x \geq \lambda - 1 \quad \dots (1)
\end{aligned}$$

$$\text{Similarly } p(x) \geq p(x-1) \Rightarrow x \leq \lambda \quad \dots (2)$$

Combining (1) and (2), we have

$$\lambda - 1 \leq x \leq \lambda$$

Hence mode of the Poisson distribution lies between $\lambda - 1$ and λ .

Case 1 : If λ is an integer then $\lambda - 1$ is also an integer. So we have two maximum values and the distribution is bimodal and the two modes are $(\lambda - 1)$ and λ .

Case 2 : If λ is not an integer, the mode of Poisson distribution is integral part of λ .

4. Recurrence Relation for the Poisson Distribution

$$\text{We have } p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\therefore p(x+1) = \frac{e^{-\lambda} \cdot \lambda^{x+1}}{(x+1)!} = \frac{\lambda}{x+1} \cdot \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{\lambda}{x+1} p(x)$$

Thus
$$p(x+1) = \left(\frac{\lambda}{x+1} \right) p(x)$$

or
$$p(x) = \frac{\lambda}{x} \cdot p(x-1)$$

which is the required relation. With this formula we can find $p(1), p(2), p(3), \dots$ if $p(0)$ is given.

Note : 1. On successive application of the formula, we get

$$p(x+1) = \frac{\lambda^{x+1}}{(x+1)!} \cdot p(0)$$

$$2. P(X \leq t) = \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^t}{t!} \right] \cdot p(0)$$

$$3. P(a \leq X \leq b) = \left[\frac{\lambda^a}{a!} + \frac{\lambda^{a+1}}{(a+1)!} + \dots + \frac{\lambda^b}{b!} \right] \cdot p(0), \text{ for all } 0 \leq a, b, a, b \in \mathbb{Z}^+$$

Properties of Poisson Distribution

1. Range of the variable is from 0 to ∞ .
2. Mean and Variance are equal.
3. Distribution gets more and more symmetrical about the mean as λ increases and tends to normal distribution, described in the next section.

SOLVED EXAMPLES

Example 1 : If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (i) exactly 3 (ii) more than 2 individuals (iii) none (iv) more than one individual suffer a bad reaction.

Solution : Given $p = 0.001$ and $n = 2000$

$$\therefore \text{Mean, } \lambda = np = 2000 (0.001) = 2$$

$$\therefore P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!} = \frac{e^{-2} \cdot 2^r}{r!} = \frac{2^r}{e^2 r!}$$

$$(i) P(3) = \frac{1}{e^2} \cdot \frac{2^3}{3!} = \frac{8}{6(2.718)^2} = 0.1804$$

$$\begin{aligned} (ii) P(\text{more than } 2) &= P(r \geq 2) = P(3) + P(4) + \dots + P(2000) \\ &= 1 - [P(0) + P(1) + P(2)] = 1 - \left[\frac{1}{e^2} + \frac{2}{e^2} + \frac{4}{2e^2} \right] \\ &= 1 - \frac{1}{e^2} (1 + 2 + 2) = 1 - \frac{5}{e^2} = 1 - 0.67667 = 0.3233 \end{aligned}$$

$$(iii) P(\text{none}) = P(0) = \frac{2^0}{e^2 0!} = \frac{1}{e^2} = 0.1353$$

$$\begin{aligned}
 \text{(iv)} \quad P(\text{more than one}) &= P(r > 1) = P(2) + P(3) + \dots + P(2000) = 1 - [P(0) + P(1)] \\
 &= 1 - \left(\frac{1}{e^2} + \frac{2}{e^2} \right) = 1 - \frac{3}{e^2} = 1 - 0.406 = 0.594
 \end{aligned}$$

Example 2 : A car-hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Calculate the proportion of days (i) on which there is no demand (ii) on which demand is refused. [JNTU 2003, (K) May 2013 (Set No. 4)]

Solution : Given mean, $\lambda = 1.5$

$$\text{We have } P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

$$\text{(i)} \quad P(\text{no demand}) = P(0) = \frac{e^{-1.5} (1.5)^0}{0!} = e^{-1.5} = 0.2231$$

Note : Number of days in a year there is no demand of car = $365 (0.2231) = 81$ days

(ii) Some demand is refused if the number of demands is more than two i.e., $r > 2$

$$\begin{aligned}
 P(\text{demand refused}) &= P(r > 2) = 1 - [P(0) + P(1) + P(2)] \\
 &= 1 - \left[e^{-1.5} + \frac{e^{-1.5} (1.5)}{1!} + \frac{e^{-1.5} (1.5)^2}{2!} \right] \\
 &= 1 - e^{-1.5} [1 + 1.5 + 1.125] \\
 &= 1 - 3.625 (e^{-1.5}) = 1 - 0.8088 = 0.1913
 \end{aligned}$$

Note : Number of days in a year when some demand is refused

$$= 365 \times 0.1913 = 69.82 = 70 \text{ days}$$

Example 3 : A hospital switch board receives an average of 4 emergency calls in a 10 minute interval. What is the probability that (i) there are at most 2 emergency calls in a 10 minute interval (ii) there are exactly 3 emergency calls in a 10 minute interval.

[JNTU 2000, (K) May 2010S (Set No. 3)]

Solution : Mean, $\lambda = (4 \text{ calls} / 10 \text{ minutes}) = 4 \text{ calls}$

$$\therefore P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-4} \cdot 4^x}{x!} = \frac{1}{e^4} \cdot \frac{4^x}{x!}$$

$$\begin{aligned}
 \text{(i)} \quad P(\text{at most 2 calls}) &= P(X \leq 2) \\
 &= P(X = 0) + P(X = 1) + P(X = 2) \\
 &= \frac{1}{e^4} + \frac{1}{e^4} \cdot 4 + \frac{1}{e^4} \cdot \frac{4^2}{2!} \\
 &= \frac{1}{e^4} (1 + 4 + 8) = 13 e^{-4} = 0.2381
 \end{aligned}$$

$$\text{(ii)} \quad P(\text{exactly 3 calls}) = P(X = 3) = \frac{1}{e^4} \cdot \frac{4^3}{3!} = \frac{32}{3} e^{-4} = 0.1954$$

Example 10 : Using Poisson's distribution, find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials.

Solution : Here $p = \frac{1}{52}$ and $n = 104$

$\therefore$ Mean of the distribution, $\lambda = np = \frac{104}{52} = 2$

$P(\text{at least once}) = P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - \frac{e^{-2}(2)^0}{0!} = 1 - 0.1353 = 0.8647$$

Example 11 : The average number of phone calls / minute coming into a switch board between 2 p.m and 4 p.m is 2.5. Determine the probability that during one particular minute there will be (i) 4 or fewer (ii) more than 6 calls. [JNTU 2006 S, 2007 (Set No. 1)]

Solution : Let x be the number of phone calls / minute coming into a switch board.

Given mean, $\lambda = 2.5$

Now the Poisson distribution is

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2.5} \cdot (2.5)^x}{x!}$$

$$(i) \quad p(x \leq 4) = p(x=0) + p(x=1) + p(x=2) + p(x=3) + p(x=4)$$

$$= e^{-2.5} \left[\frac{(2.5)^0}{0!} + \frac{(2.5)^1}{1!} + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!} + \frac{(2.5)^4}{4!} \right]$$

$$= e^{-2.5} [1 + 2.5 + 3.125 + 2.6042 + 1.6276]$$

$$= (0.080209) (10.8568) = 0.8912$$

$$(ii) \quad p(x > 6) = 1 - p(x \leq 6) = 1 - e^{-2.5} \left[\frac{(2.5)^0}{0!} + \frac{(2.5)^1}{1!} + \dots + \frac{(2.5)^6}{6!} \right]$$

$$= 1 - e^{-2.5} [1 + 2.5 + 3.125 + 2.6042 + 1.6276 + 0.8138 + 0.3391]$$

$$= 1 - (0.080209) (12.01) = 0.01416$$

Example 12 : 2% of the items of a factory are defective. The items are packed in boxes. What is the probability that there will be

(i) 2 defective items (ii) at least three defective items in a box of 100 items ?

Solution : Given $n = 100$

[JNTU 2006S, (K) Nov. 2011 (Set No. 4)]

and $p =$ The probability of defective items $= 2\% = 0.02$

$\lambda =$ Mean number of defective items in a box of 100 $= np = (0.02) (100) = 2$

Since p is small, we may use Poisson distribution. Probability of x defective items in a box of 100 is

$$P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} \cdot 2^x}{x!};$$

$$(i) \quad p(2 \text{ defective items}) = p(2) = \frac{e^{-2} \cdot 2^2}{2!} = \frac{2}{e^2} = 0.2706$$

$$\begin{aligned}
 \text{(ii) } p(\text{at least } 3) &= p(x \geq 3) = 1 - [p(x=0) + p(x=1) + p(x=2)] \\
 &= 1 - e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!} + \frac{2^2}{2!} \right] \\
 &= 1 - e^{-2} [1 + 2 + 2] = 1 - 5e^{-2} = 0.3233
 \end{aligned}$$

Example 13 : Average number of accidents on any day on a national highway is 1.8. Determine the probability that the number of accidents are (i) at least one (ii) at most one [JNTU 2004S, 2005, 2007(Set No. 3)]

Solution : Mean, $\lambda = 1.8$

$$\text{We have } P(X=x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.8} \cdot (1.8)^x}{x!}$$

$$\begin{aligned}
 \text{(i) } P(\text{at least one}) &= P(X \geq 1) = 1 - P(X=0) = 1 - e^{-1.8} = 1 - 0.1653 = 0.8347 \\
 \text{(ii) } P(\text{at most one}) &= P(X \leq 1) = P(X=0) + P(X=1) \\
 &= e^{-1.8} + e^{-1.8} (1.8) = e^{-1.8} (2.8) = 0.4628
 \end{aligned}$$

Example 14 : Average number of accidents on any day on a national highway is 1.6. Determine the probability that the number of accidents are

i) at least one ii) at most one

[JNTU 2008 (Set No. 2)]

Solution : Given Mean, $\lambda = 1.6$. We have

$$p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.6} (1.6)^x}{x!}$$

$$\begin{aligned}
 \text{(i) } P(\text{at least one}) &= P(X \geq 1) = 1 - P(X=0) \\
 &= 1 - e^{-1.6} = 1 - 0.2019 = 0.7981 \\
 \text{(ii) } P(\text{at most one}) &= P(X \leq 1) = P(X=0) + P(X=1) \\
 &= e^{-1.6} + e^{-1.6} (1.6) = e^{-1.6} (2.6) \\
 &= (0.2019) (2.6) = 0.5249
 \end{aligned}$$

Example 15 : Number of monthly breakdowns of a computer is a random variable having Poisson distribution with mean equal to 1.8. Find the probability that the computer will function for a month (i) without a breakdown (ii) with only one breakdown and (iii) with at least one. [JNTU (K) 2009 (Set No. 3)]

Solution : Given mean, $\lambda = 1.8$

$$\text{From Poisson distribution, } P(X=x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{(e^{-1.8})(1.8)^x}{x!}$$

$$\begin{aligned}
 \text{(i) } p(X=0) &= \frac{(e^{-1.8})(1.8)^0}{0!} = e^{-1.8} = 0.1653 \\
 \text{(ii) } p(X=1) &= (e^{-1.8})(1.8) = 0.2975 \\
 \text{(iii) } p(X \geq 1) &= 1 - p(X < 1) = 1 - p(X=0) = 1 - 0.1653 = 0.8347
 \end{aligned}$$

Example 24 : If X is a Poisson Variate such that $3p(x=4) = \frac{1}{2} p(x=2) + p(x=0)$, find (i) the mean of x (ii) $p(x \leq 2)$ [JNTU 2008S, (A) Nov. 2010, (H) May 2011 (Set No.1)]

Solution : (i) If X is a Poisson Variate with parameter λ , then

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, x = 0, 1, 2, \dots, \lambda > 0$$

Here we are given $3p(x=4) = \frac{1}{2} p(x=2) + p(x=0)$

$$\Rightarrow 3 \cdot \frac{e^{-\lambda} \lambda^4}{4!} = \frac{1}{2} \cdot \frac{e^{-\lambda} \cdot \lambda^2}{2!} + e^{-\lambda}$$

$$\Rightarrow \lambda^4 - 2\lambda^2 - 8 = 0 \Rightarrow (\lambda^2 - 4)(\lambda^2 + 2) = 0$$

$$\Rightarrow \lambda = \pm 2 \Rightarrow \lambda = 2 \quad (\because \lambda > 0)$$

$\therefore$ Mean of the Poisson variate i.e., $\lambda = 2$

(ii) Now we have $P(X=x) = \frac{e^{-2} \cdot 2^x}{x!}$

$$p(x \leq 2) = p(x=0) + p(x=1) + p(x=2)$$

$$= e^{-2} + \frac{1}{4} e^{-1} + \frac{e^{-2} \cdot 2^2}{2!}$$

$$= 3e^{-2} + \frac{1}{4} e^{-1} = 3(0.135) + \frac{1}{4} (0.3679)$$

$$= 0.498$$

Example 25 : If 2% of light bulbs are defective. Find

- i. At least one is defective.
- ii. Exactly 7 are defective.
- iii. $P(1 < x < 8)$ in a sample of 100

[JNTU (H) Dec. 2011 (Set No. 1)]

Solution : We are given $n = 100$

Let p = The probability of a defective bulb = 2% = 0.02

$$\begin{aligned}\therefore \lambda &= \text{Mean number of defective bulbs in a sample of 100} \\ &= np = 100 \times 0.02 = 2\end{aligned}$$

Since 'p' is small, we may use Poisson distribution.

Probability of x defective bulbs in a sample of 100 is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-2} 2^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) P(\text{At least one is defective}) = P(x \geq 1)$$

$$= 1 - P(x = 0)$$

$$= 1 - \frac{e^{-2} \cdot 2^0}{0!} = 1 - e^{-2}$$

$$= 1 - 0.1353 = 0.8647$$

$$(ii) P(\text{Exactly 7 are defective}) = P(x = 7)$$

$$= \frac{e^{-2} 2^7}{7!} = \frac{0.1353 \times 128}{5040} = 0.0034$$

$$(iii) P(1 < x < 8) = P(x = 2) + P(x = 3) + P(x = 4) + P(x = 5) + P(x = 6) + P(x = 7)$$

$$= \frac{e^{-2} \cdot 2^2}{2!} + \frac{e^{-2} \cdot 2^3}{3!} + \frac{e^{-2} \cdot 2^4}{4!} + \frac{e^{-2} \cdot 2^5}{5!} + \frac{e^{-2} \cdot 2^6}{6!} + \frac{e^{-2} \cdot 2^7}{7!}$$

$$= \frac{e^{-2}}{2} \left(4 + \frac{8}{3} + \frac{16}{12} + \frac{32}{60} + \frac{64}{360} + \frac{128}{2520} \right)$$

$$= \frac{0.1353}{2} (4 + 2.6666 + 1.3333 + 0.5333 + 0.1777 + 0.0508)$$

$$= \frac{0.1353}{2} (8.7617) = 0.593$$

✓ **Example 26 :** Given that $p(x = 2) = 9p(x = 4) + 90p(x = 6)$ for a Poisson variate X. Find :

$$(i) P(x < 2) \quad (ii) P(x > 4) \quad (iii) P(x \geq 1) \quad [\text{JNTU (H) Dec. 2011 (Set No. 2)}]$$

Solution : If X is a Poisson variate with parameter λ , then

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, x = 0, 1, 2, \dots; \lambda > 0$$

Since $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, we have

$$\frac{e^{-\lambda} \cdot \lambda^2}{2!} = 9 \cdot \frac{e^{-\lambda} \cdot \lambda^4}{4!} + 90 \cdot \frac{e^{-\lambda} \cdot \lambda^6}{6!}$$

$$= \frac{e^{-\lambda} \cdot \lambda^2}{8} [3\lambda^2 + \lambda^4]$$

$$\Rightarrow 4 = 3\lambda^2 + \lambda^4 \text{ or } \lambda^4 + 3\lambda^2 - 4 = 0 \text{ or } (\lambda^2 + 4)(\lambda^2 - 1) = 0$$

$$\therefore \lambda^2 + 4 = 0 \text{ or } \lambda^2 - 1 = 0. \text{ But } \lambda \text{ cannot be imaginary.}$$

[(or) This is a quadratic equation in λ^2

$$\therefore \lambda^2 = \frac{-3 \pm \sqrt{9+16}}{2} = \frac{-3 \pm 5}{2} = 1, -4$$

Since $\lambda > 0$, we get

$$\lambda^2 = 1 \Rightarrow \lambda = 1, \text{ taking positive sign}$$

$$\text{Hence } P(X = x) = \frac{e^{-1} \cdot 1^x}{x!}, x = 0, 1, 2, \dots$$

$$(i) P(x < 2) = P(x = 0) + P(x = 1)$$

$$= e^{-1} + e^{-1} = \frac{2}{e} = 0.7358$$

$$(ii) P(x > 4) = 1 - P(x \leq 4)$$

$$= 1 - [P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4)]$$

$$= 1 - \left[e^{-1} + e^{-1} + \frac{e^{-1}}{2} + \frac{e^{-1}}{3!} + \frac{e^{-1}}{4!} \right]$$

$$= 1 - e^{-1} \left[2 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \right] = 1 - e^{-1} \left[\frac{65}{24} \right]$$

$$= 1 - 0.9963 = 0.0037$$

$$(iii) P(x \geq 1) = 1 - p(x = 0) = 1 - e^{-1} = 1 - 0.3679 = 0.6321$$

Example 27 : If the Mean of a Poisson variable is 1.8, then find

$$(i) p(x > 1)$$

$$(ii) p(x = 5)$$

$$(iii) p(0 < x < 5) \text{ [JNTU (H) Dec. 2011 (Set No.3)]}$$

Sol. Given mean of a Poisson variate, $\lambda = 1.8$

For Poisson distribution, the probability of x successes is

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-1.8} (1.8)^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad p(x > 1) = 1 - p(x \leq 1)$$

$$= 1 - [p(x=0) + p(x=1)]$$

$$= 1 - [e^{-1.8} + e^{-1.8}(1.8)] = 1 - e^{-1.8}(1 + 1.8) = 1 - 0.4628 = 0.5372$$

$$(ii) \quad p(x=5) = \frac{e^{-1.8}(1.8)^5}{5!} = \frac{3.1234}{120} = 0.026$$

$$(iii) \quad p(0 < x < 5) = p(x=1) + p(x=2) + p(x=3) + p(x=4)$$

$$= e^{-1.8} + \frac{e^{-1.8}(1.8)^2}{2!} + \frac{e^{-1.8}(1.8)^3}{3!} + \frac{e^{-1.8}(1.8)^4}{4!}$$

$$= e^{-1.8} \left[1 + \frac{(1.8)^2}{2} + \frac{(1.8)^3}{6} + \frac{(1.8)^4}{24} \right] = e^{-1.8} [1 + 1.62 + 0.972 + 0.4374]$$

$$= e^{-1.8}(4.0294) = 0.666$$

Example 28 : If 2% of light bulbs are defective, find

(i) At least one is defective. (ii) Exactly 7 are defective (iii) $p(1 < x < 8)$

in a sample of 100

[JNTU (H) Apr. 2012 (Set No. 3)]

Solution : Given $n = 100$ and

p = The probability of defective light bulbs = 0.02

$\therefore$ Mean = λ = Mean number of defective light bulbs in a sample of 100.

$$= np = 100(0.02) = 2$$

Since p is small, we use Poisson distribution.

Probability of x defective light bulbs in a sample of 100 is

$$P(X = x) = p(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} \cdot 2^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) \quad p(\text{at least one is defective}) = p(x \geq 1) = 1 - p(x=0)$$

$$= 1 - \frac{e^{-2} \cdot 2^0}{0!} = 1 - e^{-2} = 0.8646$$

$$(ii) \quad p(\text{exactly 7 are defective}) = p(x=7)$$

$$= \frac{e^{-2} \cdot 2^7}{7!} = \frac{128 e^{-2}}{5040} = 0.0034$$

$$(iii) \quad p(1 < x < 8) = p(x=2) + p(x=3) + p(x=4) + p(x=5) + p(x=6) + p(x=7)$$

$$= e^{-2} \left[\frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \frac{2^5}{5!} + \frac{2^6}{6!} + \frac{2^7}{7!} \right]$$

$$\begin{aligned}
 &= e^{-2} \left[\frac{4}{2} + \frac{8}{6} + \frac{16}{24} + \frac{32}{120} + \frac{64}{720} + \frac{128}{5040} \right] \\
 &= e^{-2} (2 + 1.33 + 0.66 + 0.27 + 0.09 + 0.02) \\
 &= e^{-2} (4.37) = 0.59
 \end{aligned}$$

Example 29 : Poisson variable has a double mode at $x = 2$ and $x = 3$, find the maximum probability and also find $p(x \geq 2)$. [JNTU (H) Apr. 2012 (Set No. 4)]

Solution : We know that the Poisson distribution is bimodal and two modes are at the points $x = \lambda - 1$ and $x = \lambda$.

Since we are given that the two modes are at the points $x = 1$ and $x = 2$, we find that $\lambda = 2$.

$$\therefore P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!} = \frac{e^{-2} 2^x}{x!}; x = 0, 1, 2, \dots$$

$$\text{Now } P(X = 1) = 2e^{-2} \text{ and } P(X = 2) = \frac{e^{-2} \cdot 2^2}{2!} = 2e^{-2}$$

$$\text{Hence the required probability} = P(X = 1) + P(X = 2) = 2e^{-2} + 2e^{-2} = 4e^{-2} = 0.5413$$

$$\text{Now } p(x \geq 2) = 1 - p(x < 2)$$

$$= 1 - [p(x = 0) + p(x = 1) + p(x = 2)]$$

$$= 1 - [e^{-2} + 2e^{-2} + 2e^{-2}] = 1 - 5e^{-2} = 1 - 0.676$$

$$= 0.3233$$

Example 30 : A sample of 3 items is selected at random from a box containing 10 items of which 4 are defective. Find the expected number of defective items?

[JNTU (K) May 2010 (Set No. 2)]

Solution : The probability of defective is $p = \frac{4}{10} = \frac{2}{5}$

Number of items chosen, $n = 3$

The expected number of defective is $E(X) = \mu = np = 3 \left(\frac{2}{5} \right) = \frac{6}{5} = 1.2 \approx 1$

Alternative Method :

$$\text{No. of Exhaustive cases} = {}^{10}C_3 = \frac{10!}{3!7!} = 120$$

$$\text{Probability that there are no defective items} = p(x = 0) = \frac{{}^6C_3}{120} = \frac{1}{6}$$

$$\text{Probability that there is one defective item} = p(x=1) = \frac{{}^6C_2 \cdot {}^4C_1}{120} = \frac{1}{2}$$

$$\text{Probability that there are two defective items} = p(x=2) = \frac{{}^6C_1 \cdot {}^4C_2}{120} = \frac{3}{10}$$

$$\text{Probability that there are three defective items} = p(x=3) = \frac{{}^4C_3}{120} = \frac{1}{30}$$

x	0	1	2	3
$p(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

$$\therefore \text{Expected number of defective items} = E(x) = \sum p_i x_i$$

$$= 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{1}{30}$$

$$= 0 + \frac{1}{2} + \frac{3}{5} + \frac{1}{10} = \frac{12}{10} = 1.2$$

Example 31 : Wireless sets are manufactured with 25 soldered joints each. On the average 1 joint in 500 is defective. How many sets can be expected to be free from defective joints in a consignment of 10,000 sets.

[JNTU Nov. 2008 (Set No. 3)]

Solution : On average, one joint in 500 joints is defective

$$\text{Number of soldered joints} = 25 = n$$

$$\text{Probability that a joint is defective} = \frac{1}{500}$$

$$\therefore \text{Mean} = np = 25 \times \frac{1}{500} = \frac{1}{20} = 0.05$$

$$\text{Thus } \lambda = 0.05$$

$$P(\text{there are } r \text{ defective joints}) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

$$\therefore P(r=0) = e^{-\lambda} = e^{-0.05}$$

$$\text{Hence expected number of sets free of defective joints among 10000 sets} \\ = 10000 \cdot e^{-0.05} = 10000 (0.951229) = 9512.29 \approx 9512$$

Example 32 : If a Poisson variate $2P(x=0) = P(x=2)$ find the probability that

$$(i) P(x \leq 3) \quad (ii) P(2 < x \leq 5) \quad (iii) P(x \geq 3)$$

[JNTU (A) Nov. 2011 (Set No. 2)]

Solution : For Poisson distribution, the probability function is $p(x, \lambda) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$, where λ is the parameter.

$$\text{By the given condition, } 2p(x=0) = p(x=2).$$

$$\Rightarrow 2e^{-\lambda} = e^{-\lambda} \cdot \frac{\lambda^2}{2!} \therefore \lambda^2 = 4. \text{ Hence } \lambda = 2.$$

$$(i) \quad p(x \leq 3) = p(x=0) + p(x=1) + p(x=2) + p(x=3)$$

$$= e^{-2} + e^{-2} \cdot 2 + e^{-2} \cdot \frac{2^2}{2!} + e^{-2} \cdot \frac{2^3}{3!}$$

$$= e^{-2} \left(1 + 2 + 2 + \frac{4}{3} \right) = \frac{19}{3} e^{-2}$$

$$= \frac{19}{3} (0.1353) = \frac{2.5714}{3} = 0.857$$

$$(ii) \quad p(2 < x \leq 5) = p(x=3) + p(x=4) + p(x=5)$$

$$= e^{-2} \cdot \frac{2^3}{3!} + e^{-2} \cdot \frac{2^4}{4!} + e^{-2} \cdot \frac{2^5}{5!} = e^{-2} \left(\frac{8}{6} + \frac{16}{24} + \frac{32}{120} \right)$$

$$= e^{-2} \left(\frac{4}{3} + \frac{2}{3} + \frac{4}{15} \right) = e^{-2} \left(2 + \frac{4}{15} \right)$$

$$= \frac{34}{15} e^{-2} = \frac{34}{15} (0.1353) = \frac{4.6002}{15} = 0.3067$$

$$(iii) \quad p(x \geq 3) = 1 - [p(x=0) + p(x=1) + p(x=2)]$$

$$= 1 - \left[e^{-2} + 2e^{-2} + e^{-2} \cdot \frac{2^2}{2!} \right]$$

$$= 1 - (5e^{-2}) = 1 - 5(0.1353) = 1 - 0.6785$$

$$= 0.3235$$

Example 33 : An insurance agent accepts policies of 5 men all of identical age and good in health. The probability that a man of this age will be alive 30 years is $\frac{2}{3}$. Find the probability that in 30 years. (i) All five men (ii) At least one man. (iii) Almost three will be alive.

[JNTU (A) Apr. 2012 (Set No. 2)]

Solution : Probability that a man will be alive 30 years $= p = \frac{2}{3}$

Number of men = 5.

$$\therefore \lambda = \text{mean} = np = 5 \left(\frac{2}{3} \right) = \frac{10}{3}$$

$$\therefore P(x \text{ men will be alive}) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-10/3} \left(\frac{10}{3} \right)^x}{x!}; x = 0, 1, 2, \dots$$

$$(i) P(x=5) = \frac{e^{-\frac{10}{3}} \left(\frac{10}{3}\right)^5}{5!} = \frac{14.6806}{120} = 0.12$$

$$(ii) P(\text{at least one man}) = P(x \geq 1) = 1 - P(x=0)$$

$$= 1 - \frac{e^{-10/3} \left(\frac{10}{3}\right)^0}{0!} = 1 - e^{-(10/3)}$$

$$= 1 - 0.0357 = 0.9643$$

$$(iii) P(\text{almost three will be alive}) = P(x \leq 3)$$

$$= 1 - [P(x=4) + P(x=5)]$$

$$= 1 - \left[\frac{e^{-(10/3)} \left(\frac{10}{3}\right)^4}{4!} + 0.12 \right]$$

$$= 1 - [0.1835 + 0.12] = 1 - 0.3035 = 0.6965$$

Example 34 : Fit a Poisson distribution for the following data and calculate the expected frequencies

x	0	1	2	3	4
$f(x)$	109	65	22	3	1

[JNTU 2004S]

Solution : Here $N = \text{total frequency} = \sum f_i = 109 + 65 + 22 + 3 + 1 = 200$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{0 + 65 + 44 + 9 + 4}{200} = \frac{122}{200} = 0.61 \quad \therefore \lambda = 0.61$$

$\therefore$ Mean of Poisson distribution, $\lambda = 0.61$

Hence the theoretical frequencies are given by $N.P(x)$, where $x = 0, 1, 2, 3, 4$

$$\text{i.e., } 200 \cdot \frac{e^{-0.61} (0.61)^x}{x!}, \text{ where } x = 0, 1, 2, 3, 4$$

$$\text{i.e., } 200 e^{-0.61}, 200 e^{-0.61} (0.61), 200 e^{-0.61} \cdot \frac{(0.61)^2}{2!}, 200 e^{-0.61} \cdot \frac{(0.61)^3}{3!}$$

$$\text{and } 200 e^{-0.61} \cdot \frac{(0.61)^4}{4!}$$

$$\text{i.e., } 108.67, 66.29, 20.22, 4.11, 0.63$$

Since frequencies are always integers, therefore by converting them to nearest integers, we get

Observed frequency	109	65	22	3	1
Expected frequency	109	66	20	4	1

Example 35 : Fit a poisson distribution to the following data

x	0	1	2	3	4	5	Total
f	142	156	69	27	5	1	400

[JNTU (K) Nov. 2009, May 2013 (Set No. 1)]

Solution : Mean = $\frac{\sum f_i x_i}{\sum f_i} = \frac{0+156+138+81+20+5}{400} = \frac{400}{400} = 1$

$\therefore$ Mean of Poisson distribution i.e., $\lambda = 1$

Hence the theoretical frequency for x successes is given by

$N.P(x)$ where $x = 0, 1, 2, 3, 4, 5$

i.e., $400 \cdot \frac{e^{-1}(1)^x}{x!}$, where $x = 0, 1, 2, 3, 4, 5$

i.e., $400(e^{-1})$, $400(e^{-1})$, $200(e^{-1})$, $66.67(e^{-1})$, $16.67(e^{-1})$, $3.33(e^{-1})$

i.e., 147.15, 147.15, 73.58, 24.53, 6.13, 1.23

$\therefore$ The theoretical frequencies are

x	0	1	2	3	4	5
f	147	147	74	25	6	1

Example 36 : The distribution of typing mistakes committed by a typist is given below.

Assuming the distribution to be Poisson, find the expected frequencies.

x	0	1	2	3	4	5
$f(x)$	42	33	14	6	4	1

Solution : $N = \sum f_i = 100$

Mean of the distribution, $\lambda = \frac{\sum f_i x_i}{\sum f_i}$

$$= \frac{0+33+28+18+16+5}{100} = \frac{100}{100} = 1$$

By Poisson distribution, $p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-1}}{x!}$

$\therefore f(0) = N.p(0) = \frac{100}{e} = 36.79$, $f(1) = N.p(1) = \frac{100}{e} = 36.79$

$f(2) = N.p(2) = \frac{100}{2e} = 18.39$, $f(3) = N.p(3) = \frac{100}{6e} = 6.13$

$$f(4) = \frac{100}{24e} = 1.53, \quad f(5) = \frac{100}{120e} = 0.31$$

Since frequencies are always integers, by converting them to nearest integers, we

get

x	0	1	2	3	4	5
Observed frequency	42	33	14	6	4	1
Expected frequency	37	37	18	6	2	0

Note : In the above example we observe that the frequencies of first two values are same. Is it true for all the data ? Observe. Moreover, is it possible to get more than two same frequencies.

Poisson Distribution as an Approximation to Binomial Distribution

When n is large, say greater than 30, and p is quite small, say less than 0.1, the Binomial distribution can be approximated by the Poisson distribution with mean np and variance also np .

We know that the variance of the Binomial distribution is npq or $np(1-p)$, so that when p approaches to zero, then q or $(1-p)$ will approach to 1. That is why the variance npq of the Binomial distribution is approximated by np , in the Poisson distribution.

Relationship with Normal Distribution

When n is very large, say >30 , the Poisson distribution is well approximated by the Normal distribution, discussed in the next section.

REVIEW QUESTIONS

1. Define Poisson distribution. List the properties of Poisson distribution.
2. Compute the mean and variance of a random variable X which follow a Poisson distribution $P(x, \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, \lambda \geq 0, x = 0, 1, \dots$
 (or) Show that the mean and variance are equal for Poisson distribution.
 (or) Derive mean and variance of a Poisson distribution. [JNTU (A) Dec. 2014]
3. Prove that the Poisson distribution can be approximated by Binomial Distribution.
 [JNTU (K) Nov. 2011 (Set No. 3)]
 (or) Prove that Poisson distribution is limiting case of Binomial distribution.
 [JNTU (K) May 2010S (Set No. 1)]
4. Write the formula of mean and variance of a Poisson distribution.

EXERCISE 7(B)

1. On an average, 2 vehicles pass by a road per minute. Find the probability of 0, 1, 2, 3, 4, 5 vehicles per minute.
2. The fraction defective in a bulk is 4%. Determine the probability of not more than one defective in a sample of size 10.
3. If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs, there are 5 defective bulbs.
4. Find the probability of finding at most 5 defective fuses in a box of 200 fuses if the experience shows that 2% of such fuses will be found defective.
5. Suppose that a book of 200 pages contains 20 printing mistakes. Assume that these errors are randomly distributed throughout the book and x , the number of errors per page has a poisson distribution. Find the probability that 30 pages selected at random will be free of errors.
6. A manufacturer known that the razor blades he makes contain on an average 0.5% of defectives. He packs them in packets of 5. What is the probability that a packet picked at random will contain 3 or more faulty blades.
7. Suppose 3% of bolts made by a machine are defective, the defects occurring at random during production. If bolts are packed 50 per box, find the probability that a given box will contain 5 defectives.
8. Assume that the probability of an individual coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. What is the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.
9. In a city, 6% of all drivers get at least one parking ticket per year. Determine the probabilities that among 80 drivers
 - (i) 4 drivers will get at least one parking ticket in a year.
 - (ii) at least 3 drivers will get at least one parking ticket in a year.
10. Using Poisson distribution find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials.
11. The number of arrivals of customers during any day follows Poisson distribution with mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2 ?
12. An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year ?
13. What is the probability that an ace of spade will not be drawn even once from a deck of cards in 100 consecutive trials.
14. Out of 100 balls 50 are red and the rest white. If 60 balls are picked up at random, what is the probability of picking up (i) 3 red balls (ii) not more than 3 red balls in the sample.

15. A person receives an average of 4 telephone calls a day. Assuming Poisson distribution what is the probability that on a given day he will receive (i) no telephone call (ii) exactly only call (iii) at least two telephone calls.
16. The random variable X has a Poisson distribution. If $P(X = 1) = 0.01487$ and $P(X = 2) = 0.04461$, then find $P(X = 3)$.
17. If X is a Poisson variate such that $P(X = 0) = P(X = 2) + 3 P(X = 4)$, find (i) the mean of X (ii) $P(X \leq 2)$.
[JNTU 2004S (Set No. 3)]
18. Using recurrence formula find the probabilities when $x = 0, 1, 2, 3, 4$ and 5 , if the mean of Poisson distribution is 3 .
[JNTU 2005 (Set No. 4)]
19. Suppose that X has a Poisson distribution. If $P(X = 2) = \frac{2}{3} P(X = 1)$, find (i) $P(X = 0)$ and (ii) $P(X = 3)$.
20. If X and Y are independent Poisson variates such that $P(X = 1) = P(X = 2)$ and $P(Y = 2) = P(Y = 3)$. Find the variance of (i) X (ii) Y (iii) $X - 2Y$.
21. If X follows a poisson law such that $2P(X = 0) = P(X = 2)$, find (i) $p(X \leq 3)$ (ii) $p(2 < X \leq 5)$ (iii) $p(X \geq 3)$.
[JNTU (A) Nov. 2011 (Set No. 2)]
22. If X is a poisson variate such that $P(X = 2) = 9P(X = 4) + 90 P(X = 6)$, find the mean and variance of X .
[JNTU (K) May 2012 (Set No. 1)]
23. If X is a poisson variate such that $P(X = 2) = 45 P(X = 6) - 3 \cdot P(X = 4)$, find (i) $P(X \geq 1)$ (ii) $P(X < 2)$.
24. A Poisson distribution has a double mode at $x = 1$ and $x = 2$. What is the probability that x will have one or the other of these two values?
[JNTU (H) May 2012 (Set No. 4)]
25. If X is a Poisson variate such that $P(X = 0) = P(X = 1) = k$, determine k .

26. Fit a Poisson distribution for the following data and calculate the expected frequencies.

x	0	1	2	3	4	5	6	7	8
f	56	156	132	92	37	22	4	0	1

27. The distribution of typing mistakes committed by a typist is given below assuming the distribution to be Poisson, find the expected frequencies

x	0	1	2	3	4	5
$f(x)$	125	95	49	20	8	3

28. In 1000 sets of trials for an event of small probability the frequencies f of the numbers of x of successes are

x	0	1	2	3	4	5	6	7
f	305	365	210	80	28	9	2	1

find the expected frequencies.

- (OR) Fit a Poisson distribution to the following data

x	0	1	2	3	4	5	6	7
f	305	365	210	80	28	9	2	1

[JNTU (K) May 2010S (Set No. 4)]

29. If 0.8 % of the fuses delivered to an arsenal are defective, use the Poisson approximation to determine the probability that 4 fuses will be defective in a random sample of 400.

[JNTU (A) Dec. 2018]

30. At a checkout counter customers arrive at an average of 1.5 per minute. Find the probabilities that in any given minute of time : (i) at most 4 will arrive. (ii) Exactly 4 will arrive (iii) At least 4 will arrive.

[JNTU (A) June 2018]

ANSWERS

1. 0.1353, 0.2706, 0.2706, 0.1804, 0.0902, 0.0361
2. 0.9384
3. 0.1008
4. 0.7845
5. 0.05
6. 0.000002555
7. 0.01412
8. 0.08
9. (i) 4.8 (ii) 0.8575
10. 0.865
11. 4.994×10^{-4}
12. 0.9998
13. 0.15
14. (i) 0.2241 (ii) 0.5976
16. 0.08922
17. (i) 1.21 (ii) 0.8773
18. 0.147, 0.2205, 0.2205, 0.1653, 0.099, 0.0495
19. (i) 0.264
20. (i) 2 (ii) 3 (iii) 4
21. (i) 0.0861 (ii) 0.0308 (iii) 0.32
22. 1, 1
23. (i) 0.864 (ii) 0.408
24. 0.542
25. $1/e$
26. 70, 137, 135, 89, 44, 17, 6, 20
27. 110, 110, 55, 18, 5, 1
28. 301, 361, 217, 87, 26, 6, 1, 0